

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date	19 September 2026
Team ID	Shaik Irshad - 238W1A4256
Project Name	Automated Network Request Management In ServiceNow
Maximum Marks	4 Marks

Technical Architecture

The deliverable below includes the technical architecture diagram for the Network Request implementation, followed by the components/technologies (Table-1) and the application characteristics (Table-2).

The diagram shows the actual end-to-end flow that was implemented: the Requester submits the catalog item through the Service Portal, ServiceNow's own automation (Catalog Variables, UI Policy, Variable Set and Flow Designer) processes the request, creates the backend record, sends the notification, routes it for approval, and finally updates and exposes the Approved/Rejected status back to the requester.

Network Request - Technical Flow

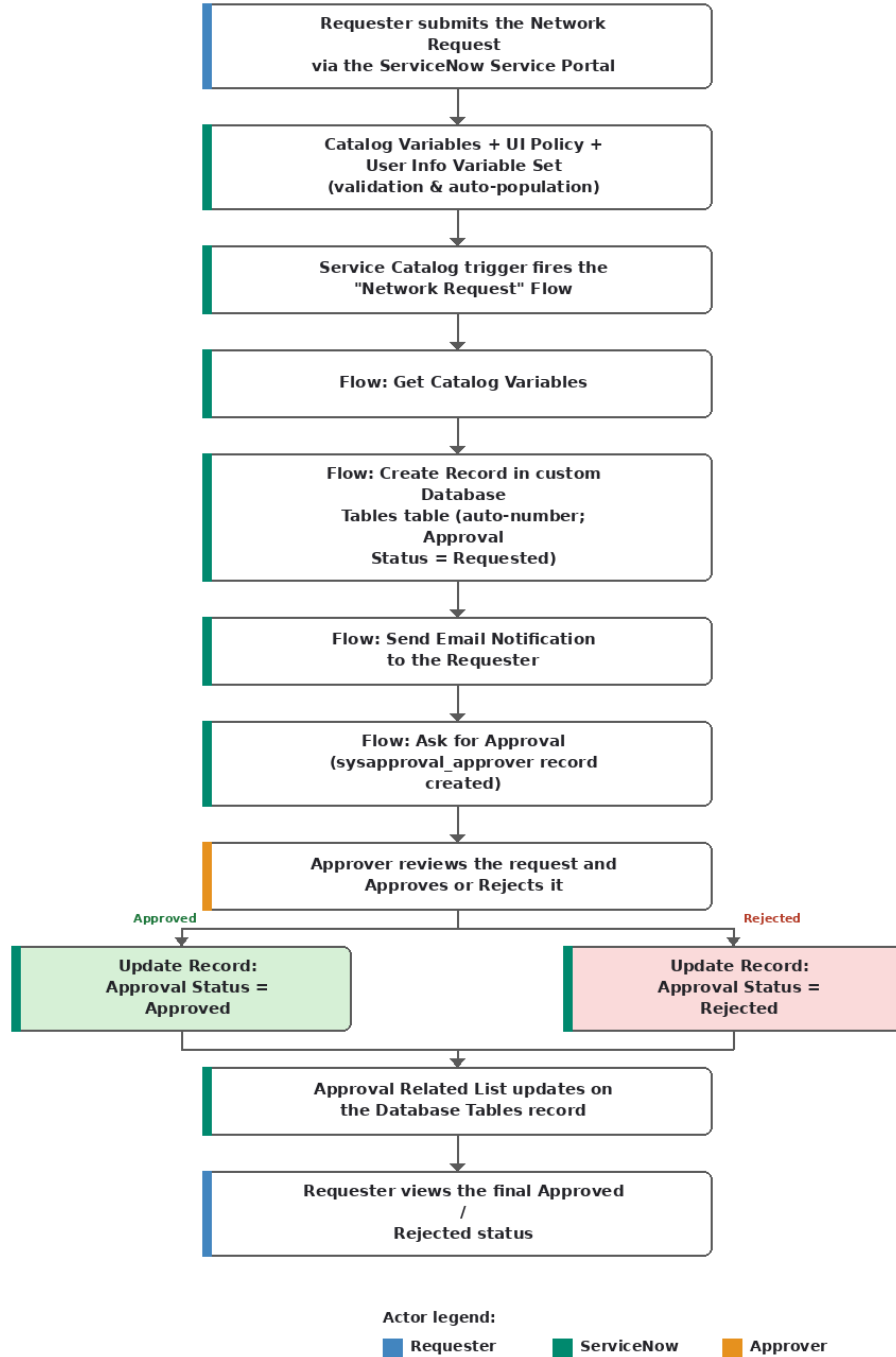


Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	How the requester interacts with the application	ServiceNow Service Portal - "Network Request" Catalog Item

S.No	Component	Description	Technology
2	Application Logic-1	Captures request details and validates/auto-populates them before submission	Catalog Variables, Catalog UI Policy, User Information Variable Set (auto-population)
3	Application Logic-2	Orchestrates backend record creation, notification and approval end to end	ServiceNow Flow Designer ("Network Request" flow, Service Catalog trigger)
4	Application Logic-3	Routes the created request for approval and evaluates the outcome	ServiceNow native Approval Engine (Ask for Approval action, sysapproval approver)
5	Database	Stores submitted network-request data as structured records	Custom "Database Tables" table created in the ServiceNow scoped application
6	Cloud Database	Underlying data store for the ServiceNow instance	ServiceNow Platform multi-instance cloud database (native to the ServiceNow SaaS platform; no separate external database service is used)
7	File Storage	File/attachment storage requirements	Not implemented — the current form does not require file attachments
8	External API-1	Purpose of an external API consumed by the application	Not implemented in this project (see Future Enhancements)
9	External API-2	Purpose of an external API consumed by the application	Not implemented in this project (see Future Enhancements)
10	Machine Learning Model	Purpose of a machine-learning model used by the application	Not applicable / not implemented in this project
11	Infrastructure (Server / Cloud)	Where the application is deployed	ServiceNow multi-tenant cloud instance (scoped application); deployment and hosting are fully managed by the ServiceNow platform — no separate server was provisioned by the team

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	The Service Portal's widget framework is bundled and managed by the ServiceNow platform itself; no additional third-party open-source framework was introduced by the project	ServiceNow Service Portal (built on AngularJS 1.x, provided natively by the platform)
2	Security Implementations	Access to the custom table and catalog item relies on ServiceNow's platform-level security rather than a custom-built framework	Default ServiceNow ACLs (role-based access control) auto-generated for the table, plus platform-standard HTTPS/TLS in transit
3	Scalable Architecture	The solution is a configuration on top of ServiceNow's native multi-instance SaaS architecture, which scales without any custom-built scaling logic	ServiceNow Now Platform (multi-instance cloud architecture, managed by ServiceNow)
4	Availability	Uptime, failover and load balancing are provided by the ServiceNow cloud hosting itself; the project did not stand up its own servers	ServiceNow SLA-backed cloud hosting (platform-managed high availability)
5	Performance	Flow Designer executes asynchronously so the requester's submission is not blocked while backend processing, email and approval creation complete	ServiceNow Flow Designer asynchronous execution; standard Service Portal client-side caching

References

<https://www.servicenow.com/docs/>

<https://developer.servicenow.com/dev.do>

<https://docs.servicenow.com/bundle/latest-release-notes/page/release-notes/family-planning/flow-designer.html>

<https://c4model.com/>